pDeep learning is a subset of machine learning that focuses on algorithms inspired by the structure and function of the brain, known as artificial neural networks. Unlike traditional machine learning methods that rely heavily on manual feature engineering, deep learning models automatically learn hierarchical feature representations from raw data. This capability has driven breakthroughs across many domains, including computer vision, natural language processing, speech recognition, and autonomous systems. At the core of deep learning are layers of interconnected neurons that transform input data through successive nonlinear operations, allowing models to capture increasingly abstract patterns. Training these networks typically involves large datasets and optimization techniques such as stochastic gradient descent, combined with backpropagation to update weights. Advances in hardware, especially GPUs and specialized accelerators, along with improvements in architectures—such as convolutional neural networks for images and recurrent or transformer-based models for sequential data—have significantly expanded the practical utility of deep learning. Despite its successes, deep learning also faces challenges: it can require substantial computational resources, large labeled datasets, and careful tuning to avoid overfitting. Interpretability and robustness remain active research areas as practitioners seek models that are not only accurate but also explainable and reliable in real-world settings. Overall, deep learning represents a powerful paradigm for extracting complex patterns from data, enabling applications that were previously infeasible and continuing to shape the future of artificial intelligence.